TOUFIK SOUADIA

BACKEND ENGINEER | NODE.JS · NESTJS · TYPESCRIPT | POSTGRESQL · REDIS · ELASTICSEARCH

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PROFESSIONAL PROFILE

Backend Engineer with 5+ years of professional experience building REST APIs, ERP systems and production web applications. Core technologies include Node.js, NestJS, TypeScript, PostgreSQL, Redis and Elasticsearch, complemented by Laravel, React and Docker-based delivery. Experienced across requirements analysis, API development, database design, authentication, caching, search, deployment and maintenance. PhD researcher in privacy-preserving federated learning for healthcare; winner of the 2025 national Linux competition.

TECHNICAL SKILLS

Backend & APIs: Node.js, NestJS, Express.js, TypeScript, Laravel (PHP), REST APIs, WebSockets, JWT/OAuth2, Odoo, ERPNext

Databases, cache & search: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, TypeORM

Infrastructure, queues & monitoring: Docker, CI/CD, Linux, Nginx, RabbitMQ, BullMQ, Grafana, Git/GitHub, Jest, PHPUnit

Frontend & desktop: React, Tailwind CSS, Inertia.js, JavaScript (ES6+), Electron

AI & research: Python, PyTorch, federated and split learning, differential privacy

PROFESSIONAL EXPERIENCE

Backend Developer | DigiBooking — May 2026 – Present

Bir Mourad Raïs, Algiers · Ride-hailing app · NestJS, TypeScript, PostgreSQL, TypeORM, Redis, Elasticsearch, RabbitMQ, BullMQ

- Designed modular REST APIs for the ride lifecycle: requests, driver assignment, status tracking and cancellation.
- Built real-time communication between passengers, drivers and the platform over WebSockets.
- Moved asynchronous work to RabbitMQ messaging and BullMQ background jobs, so long-running tasks never block API responses.
- Implemented JWT/OAuth2 authentication and role-based access for passengers, drivers and administrators.
- Designed PostgreSQL schemas and migrations, optimized queries, and added Redis caching to reduce database load under heavy traffic; built Elasticsearch search with custom scoring.
- Monitored service health and performance with Grafana dashboards.

Freelance Web Developer — Jun 2025 – May 2026

Algiers · Laravel, React, PostgreSQL

- **Kiva Star** (part-time, Jun 2025 – Feb 2026): built a custom industrial ERP, now in production, that replaced the paid ERP software the company previously used; covers inventory, sales, suppliers, invoicing and reporting.
- Continued freelance web development for clients until joining DigiBooking.

Web Developer | DevTeam — Mar 2022 – Feb 2025

Kouba, Algiers · PHP, Laravel, Node.js, Express.js, MySQL, PostgreSQL, React, Tailwind CSS, Redis, Docker, CI/CD, Nginx

- Built a budget management system for wilaya (provincial) administrations.
- Delivered client projects end-to-end, from requirements analysis and design through development, testing, deployment and maintenance, collaborating through Git and code review.
- Built REST APIs with Laravel and Node.js/Express, including authentication and role management, and developed React/Tailwind interfaces.
- Designed MySQL and PostgreSQL databases, introduced Redis caching for high-use APIs, containerized applications with Docker, and set up CI/CD deployments to Linux servers with Nginx.

ERP Developer | ONPS – National Office of School Publications — Sep 2021 – Jun 2022

El Achour, Algiers · Odoo, ERPNext, Python

- Developed a custom ERP supporting the production pipeline of school publications.
- Built inventory workflows for stock movements, stock counts and traceability, alongside production-monitoring dashboards.
- Delivered the institutional website www.onps.dz, with a publication catalogue and document management, used nationwide across all wilayas.

EARLIER EXPERIENCE Developer Intern, CERIST, Ben Aknoun (Apr – May 2019): cybersecurity research project involving custom honeypots and a P2P architecture for distributed deployment and attack-data sharing. | Web Developer Intern, Glaxiom, Bordj Kiffan (Feb – Mar 2019): HTML, CSS and JavaScript frontend development; team Git workflows.

TOUFIK SOUADIA / BACKEND ENGINEER

Research, selected projects and academic background

RESEARCH & ACADEMIC EXPERIENCE

Health Monitoring with Federated Learning (IoMT) 2021 – Present

- PhD research on privacy-preserving healthcare monitoring, with local model training on medical devices so sensitive raw data is never centralized (HIPAA/GDPR).
- Multimodal fall and anomaly detection using ST-GCN and 1D-CNN on skeleton and IMU sensor data, plus adaptive sensor fusion (QCGF) that stays resilient to faulty sensors.
- Federated optimization with FedProx and secure aggregation with SecAgg across multiple sites.
- University teaching and supervision of final-year Bachelor's and Master's projects.

RESEARCH FOCUS IoMT · human activity recognition · privacy-preserving machine learning · multimodal sensor fusion · secure aggregation

SELECTED PROJECTS

Algerian Linux System | 2025

Winner of the 2025 national Linux competition (Saïda). Built a complete system from Linux From Scratch and implemented two Rust kernel drivers for USB devices.

Linux From Scratch · Rust · USB kernel drivers

Custom ERP – Desktop Edition

Customized and extended ERPNext with additional business modules covering stock, production, finance, sales, purchasing and HR.

ERPNext · ERP customization · Desktop edition

Wifi Djelfa – E-commerce

Online store for computer components, laptops and accessories, with a custom PC builder, order tracking, delivery rates by wilaya, wishlist and cart; available in Arabic and French.

E-commerce · PC builder · Order tracking

Dar Jadida – Real Estate Portal

Multilingual real-estate portal with property search and discovery.

Web platform · Search · Multilingual content

EDUCATION

PhD in Computer Science (in progress) Privacy-preserving federated learning for connected healthcare

Master's in Computer Science USTHB, Algiers | 2018 – 2020

Bachelor's in Computer Science USTHB, Algiers | 2015 – 2018

CERTIFICATIONS & LANGUAGES

NVIDIA DLI – Fundamentals of Deep Learning (2024) | Cisco Certified Network Associate (CCNA), 2022

Arabic: Native | French: Fluent | English: Intermediate